

Logbook: Flood origin analysis (Topic 3)

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This logbook documents the workflow and analysis carried out to investigate the origin and dynamics of flooding in Rue Gray, Brussels, which experienced multiple flood events during the summer of 2024. The study combines citizen-reported flood observations with hydrological data to better understand when, where, and why flooding occurs in the area. In addition, a SWMM model and a groundwater model were developed to simulate the system behavior and support the identification of the main flood sources.

Key stakeholders involved in this project include citizens, EGEB, the local commune, and water management authorities (Hydria/VIVAQUA).

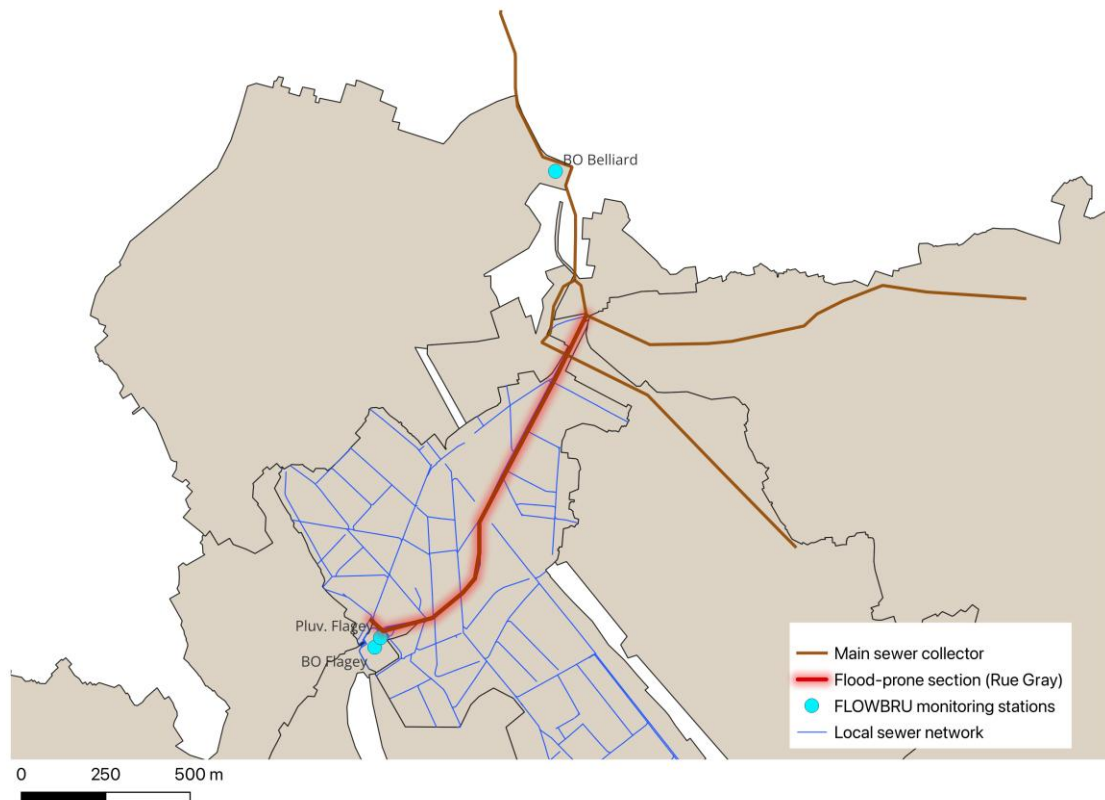


Figure 1 Study area around Rue Gray (Brussels) showing the flood-prone section, sewer network, and nearby FLOWBRU monitoring stations.

1. Analysis of citizen flood observations and FLOWBRU data analysis

In this project, we analyzed citizen-reported flood observations from Rue Gray and combined them with rainfall data to better understand flood patterns. The survey responses were first collected, organized, and cleaned into a structured dataset.

The addresses provided in the survey were georeferenced in QGIS and converted into spatial coordinates. Based on these, we created point layers representing individual flood observations. To gain more detailed insights, the data were then divided into separate datasets for each flood event. For each event, we generated heat maps using the QGIS Heatmap tool, which allowed us to visualize the spatial concentration of reported flooding and identify flood-prone areas.

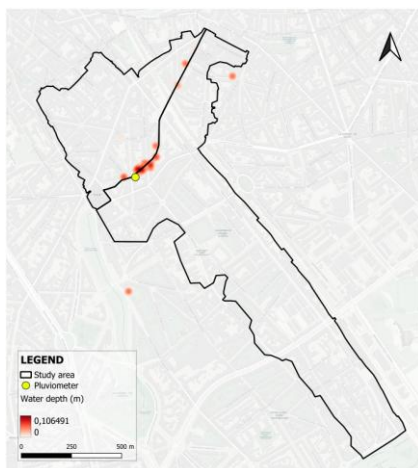


Figure 2 Heatmap 1. event (9.7.2024)

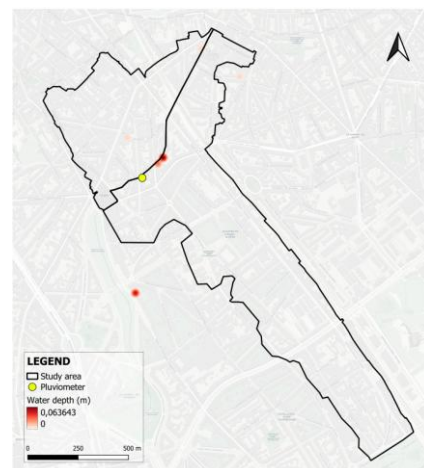


Figure 3 Heatmap 2. event (12.7.2024)

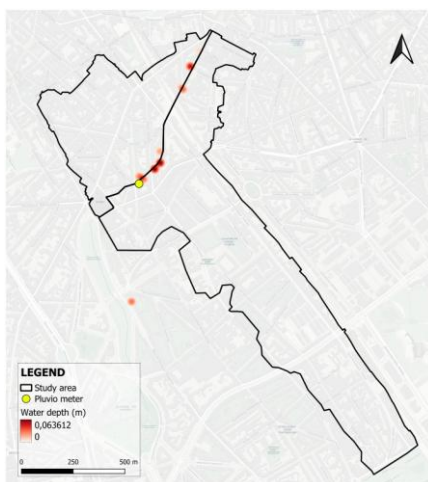


Figure 4 Heatmap 3. event (1.8.2024)

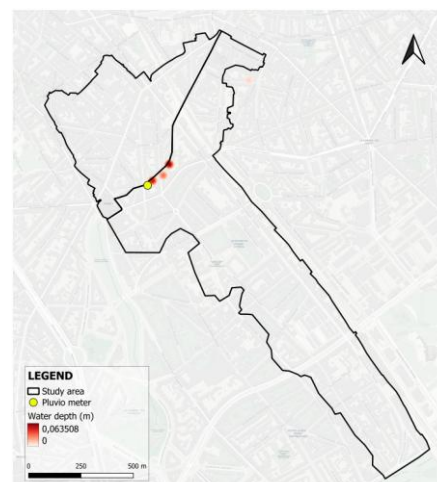


Figure 5 Heatmap 4. event (14.8.2024)

In parallel, high-resolution rainfall data were retrieved from the Flowbru (Hydria) platform, with a temporal resolution of 5 minutes. For each flood event, only the periods with significant rainfall intensity were selected and visualized in the time series to focus on the most relevant intervals.

The rainfall data were then compared with citizen observations by overlaying the reported flood timestamps onto the rainfall graphs. By cross-checking these datasets, it was observed that several citizens reported flooding immediately following peaks in rainfall intensity. This rapid response suggests that the flooding was primarily driven by surface runoff or sewer surcharge (“brown water”), rather than groundwater rise, which typically exhibits a delayed response.

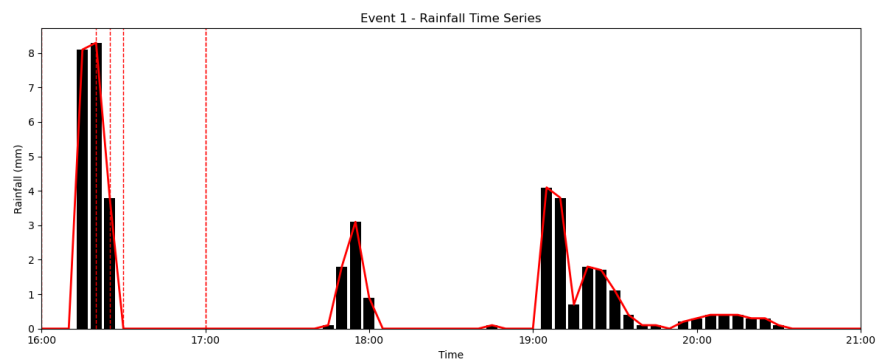


Figure 6 Rainfall Time Series 1. event (9.7.2024)

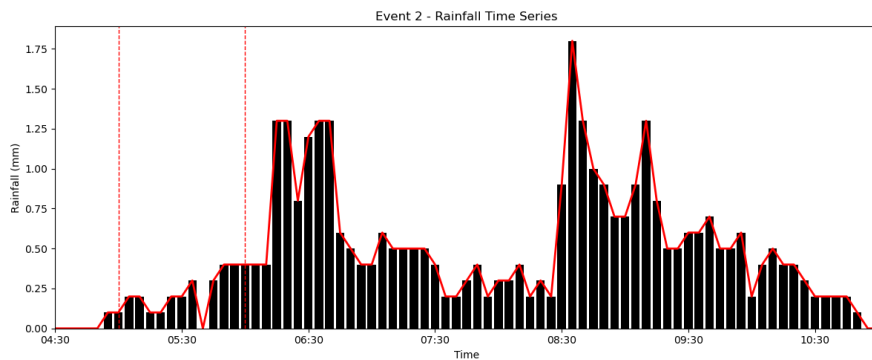


Figure 7 Rainfall Time Series 2. event (14.7.2024)

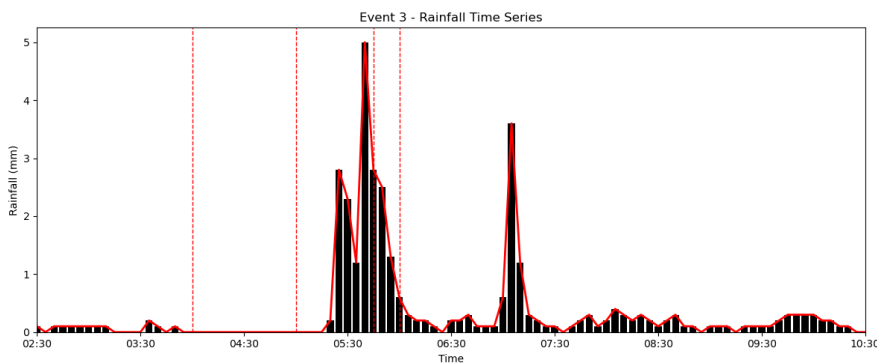


Figure 8 Rainfall Time Series 3. event (1.8.2024)

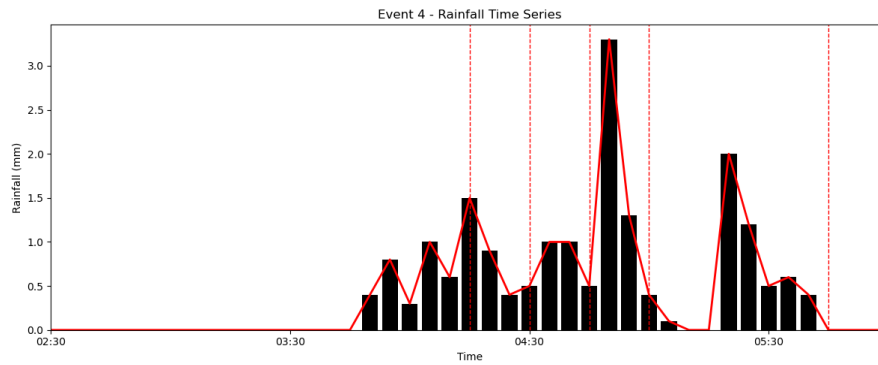


Figure 9 Rainfall Time Series 4. event (14.8.2024)

Overall, this approach helped us identify flood-prone areas and better understand how rainfall events relate to reported flooding. By combining spatial and temporal data, we were able to see when and where flooding occurred, and how quickly it followed rainfall. This provided useful insights into the likely causes of flooding and supported a more informed interpretation of the flood dynamics in the study area.

2. Develop a local SWMM model

I) Detailed Sub – Catchment area made for SWMM model

Steps taken:

1. Data Collection: Gather the Digital Elevation Model (DEM), land cover, building, and boundary layers in a matching coordinate system.
2. Clipping: Use the Clip tool to trim all vector and raster data strictly to your study area limits.
3. Land Cover Setup: Assign unique IDs and specific colors to each land cover category in the attribute table.
4. Building Removal: Use the Difference tool to subtract building footprints from the land cover layer to isolate open areas.
5. Area Calculation: Create a new field in the attribute table and use the \$area function to measure each land cover type.
6. DEM Cleaning: Process the DEM with Fill Sinks to remove depressions, then generate Flow Direction and Accumulation rasters.
7. Stream Extraction: Define a threshold on the accumulation raster to identify and create a digital stream network.
8. Pour Point Placement: Mark the outlet points where water leaves the subcatchment and snap them precisely to the streams.
9. Delineation: Run the Watershed tool using your flow direction raster and pour points to create the subcatchment boundaries.
10. Overlay Analysis: Use the Intersection tool to combine the subcatchment boundaries with the land cover data.
11. Final Reporting: Summarize the total area of each land cover type within each subcatchment for your final map.

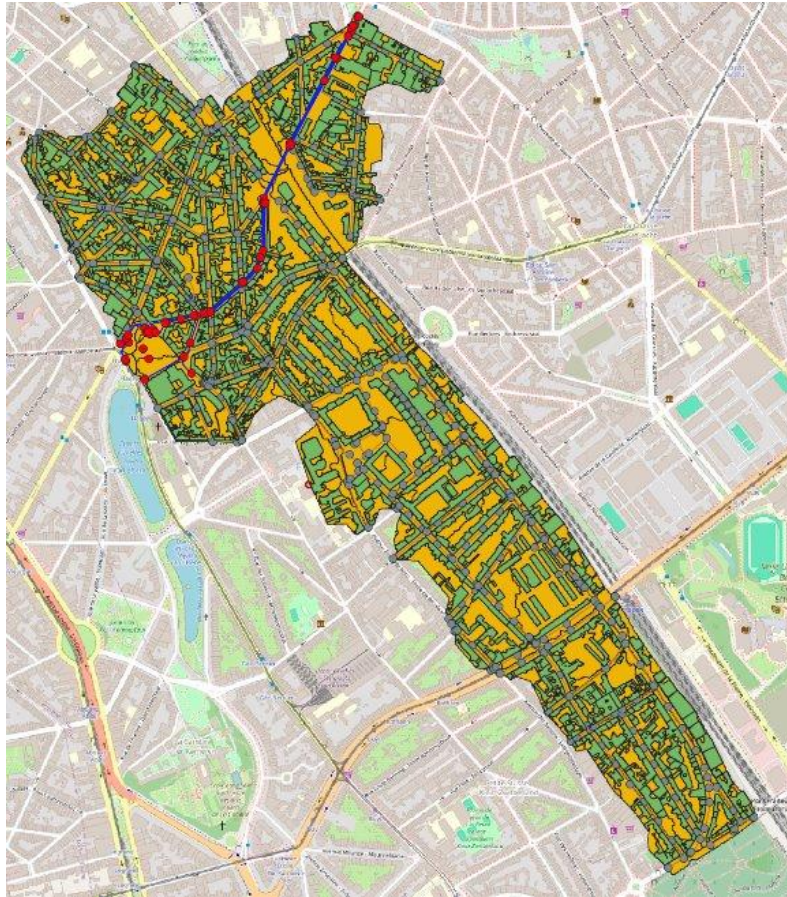


Figure 10 Sub-catchment Area

II) Data calculation for SWMM model

1. Junctions

- Invert elevation determined from the lowest connected pipe end (offset = 0).
- Offsets for other pipes assigned based on their pipe end elevations.
- For known areas (e.g., Rue Gray), invert elevations were directly available.
- Junction coordinates identified from pipe endpoint locations.
- Maximum depth = surface elevation – invert elevation.
- Initial depth, surcharge depth, and ponding depth assumed zero.

2. Conduits (Pipes)

- Most pipe properties (shape, length, elevations, geometry) were available.
- Missing data estimated using the continuity principle (smooth connection between pipes).
- Pipe ends adjusted to ensure consistent elevation and dimensions.
- Pipes initially assumed to be dry (no initial flow).

3. Sub-catchments

- Defined as rooftops and surface areas.
- Slopes obtained from GRASS GIS catchment analysis.
- Rooftops assumed 100% impervious.
- Impervious/pervious ratio derived from land use raster data.
- Green areas (grass, trees, fields) considered pervious.
- Built surfaces (roads, concrete) considered impervious.
- Areas calculated using GIS-based sub-catchment polygons.
- Flow length estimated as:
 - Rooftops → $\sqrt{\text{area}}$
 - Surfaces → longest flow path approximation

III) SWMM Model Integration and Simulation Workflow

1. Network Integration: We merged the unconnected sewer networks from two specific subcatchments into the main system.
2. Data Formatting: Using the existing main model as a base, we created a new .inp file with the required junctions and nodes by copying the features as required by SWMM format with the help of notepad+.
3. Subcatchment Consolidation: A separate .inp file was generated for the subcatchments and then merged with the junction and node data to create one unified model. This was completed by copying and pasting the new features into the subcatchment inp file in correct SWMM formats using Notepad++ to ensure data integrity.
4. Precipitation Setup: Rainfall data for gauges P14 and P11 was downloaded from Hydria; the original .tsv files were converted to .dat format for SWMM compatibility.
5. Inflow Calculation: Data from collectors C10 and C11 were integrated to account for inflows from neglected sections, simplifying the network logic.
6. Hydraulic Connections: We established connections between main three pipes at various locations by introducing weirs into the pathways to manage flow transitions.
7. Event Analysis: Once the single functional model was generated, it was simulated and analyzed across four distinct rainfall events.

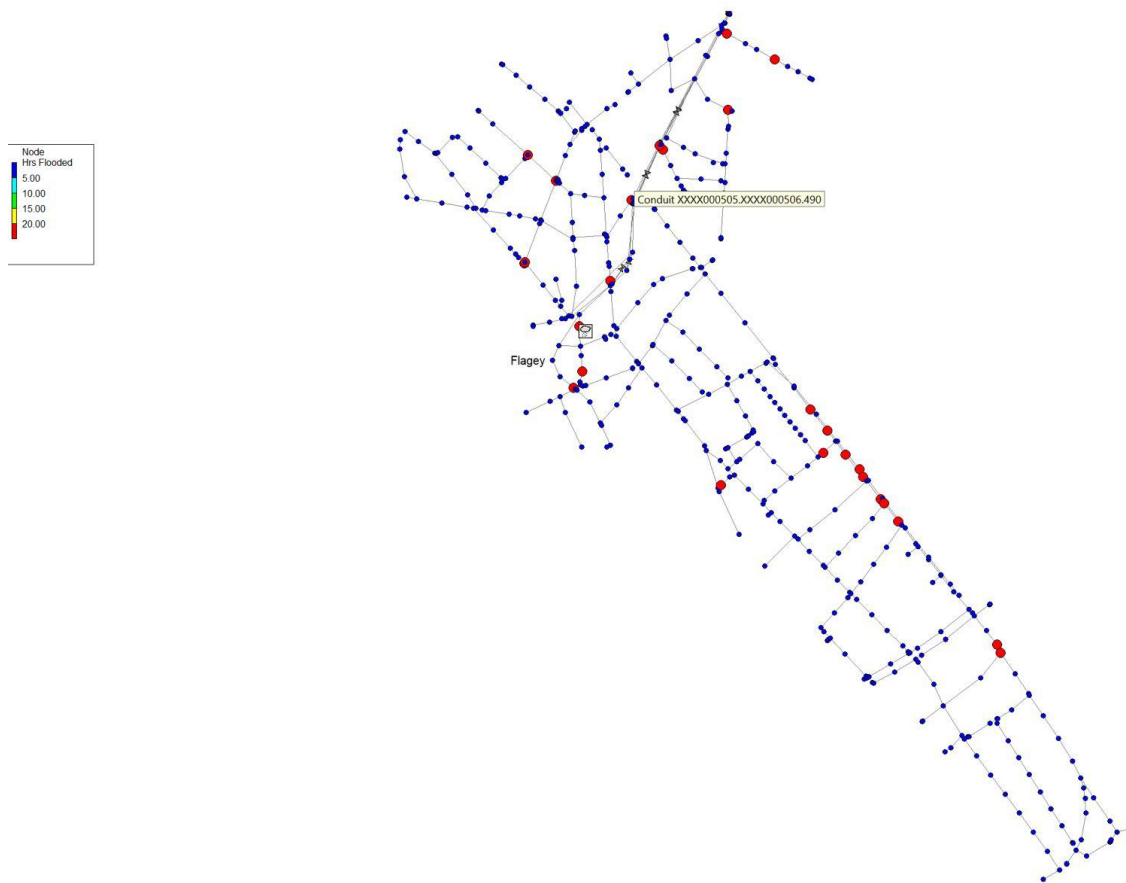


Figure 11 SWMM model showing node flooding at different locations

Results

The SWMM model was simulated using the rainfall event that occurred on 9 July 2024. This event was selected as a representative case to evaluate the model's performance under real conditions.

Following the simulation, the node flooding results were analyzed. The model identified multiple locations where flooding occurred within the drainage network.

These simulated flooding locations were then compared with citizen-reported observations from the same rainfall event. A strong agreement was found between the model results and the observed flood-prone areas.

This correspondence indicates that the model can reasonably represent the real system behavior. Therefore, the model can be considered successfully validated for this event and is suitable for further analysis.

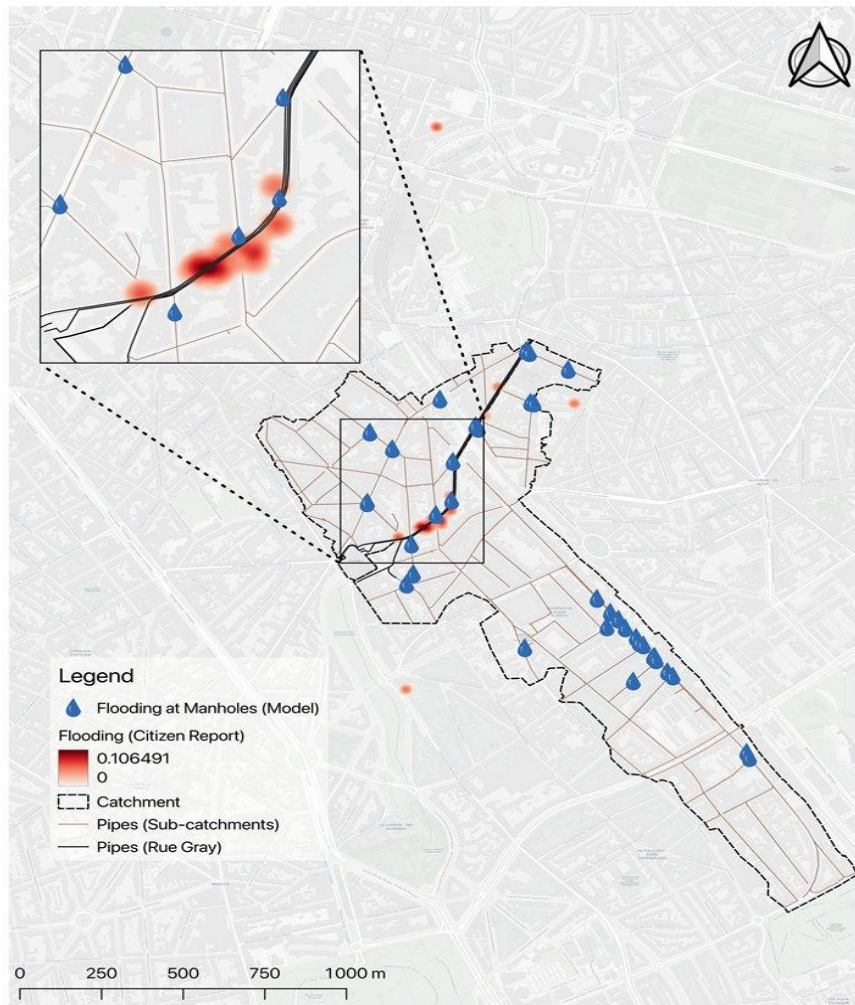


Figure 12 Validation of the SWMM model with citizen observations

3. Identify hydrological fluxes and dynamics

The geology is characterised by a confined sand bank called “Sand of Brussels” and a clay layer, that corresponds to the bottom of the aquifer, below the sand bank, that takes places on the surface near the river section. The hydraulic conductivity (K) of the Sand of Brussels is 17.28 m/day, according to the value indicated by BRUSSELS PHREATIC SYSTEM MODEL - VERSION 1.0. The boundary conditions set for the southern and eastern sides, corresponding to the hills of the valley, are no flow boundaries, while the river is set as river package, which value of the top is calculated as the digital elevation model minus 3 meters, and the bottom is the digital elevation model minus 4 meters. The presence of a railway, that is considered as an impervious barrier for the water flow, while the so-called bottle neck starts from a bridge (no flow boundary) few meters in southern direction of Rue Grey respect to the railway. The figures of the study area, which backgrounds are made by ESRI and OSM (Figure 13, Figure 14) and the Cross-Sections, made on Brugeotool are reported as follows (Figure 15, Figure 16, Figure 17, Figure 18, Figure 19, Figure 20, Figure 21, Figure 22, Figure 23, Figure 24, Figure 25, Figure 26):

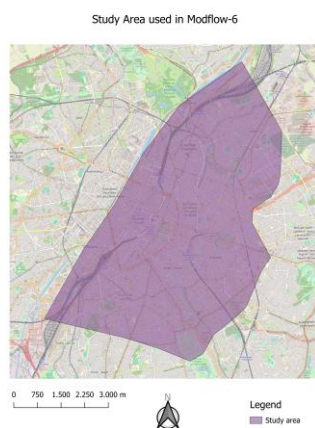


Figure 13 Study area for the GW model with street map (ESRI Gray)

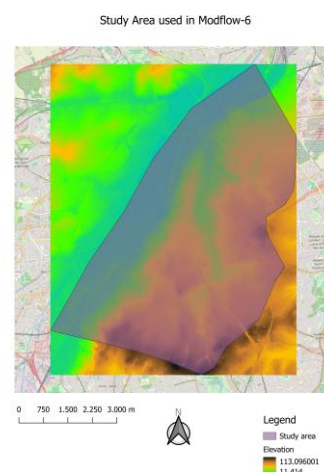


Figure 14 Study area for the GW model with elevation (OSM standard)



Figure 15 Cross-section of the upstream study area seen from above (Brugeotool)

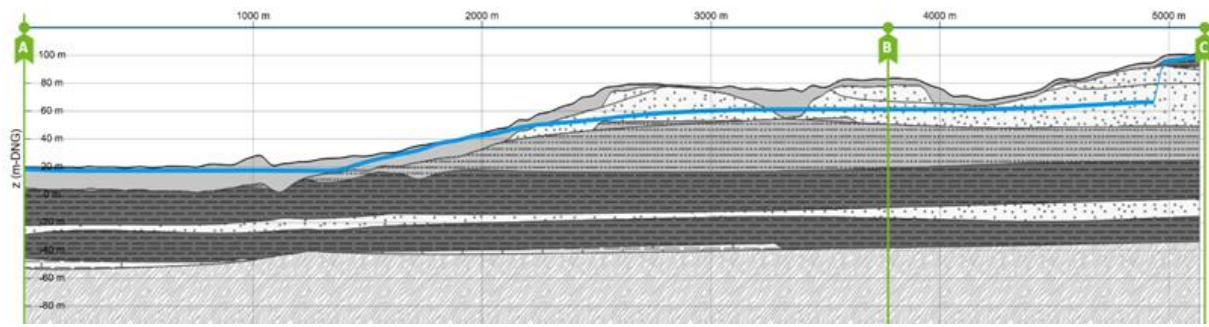


Figure 16 Cross-section of the hydrological condition of the upstream study area (Brugeotool)

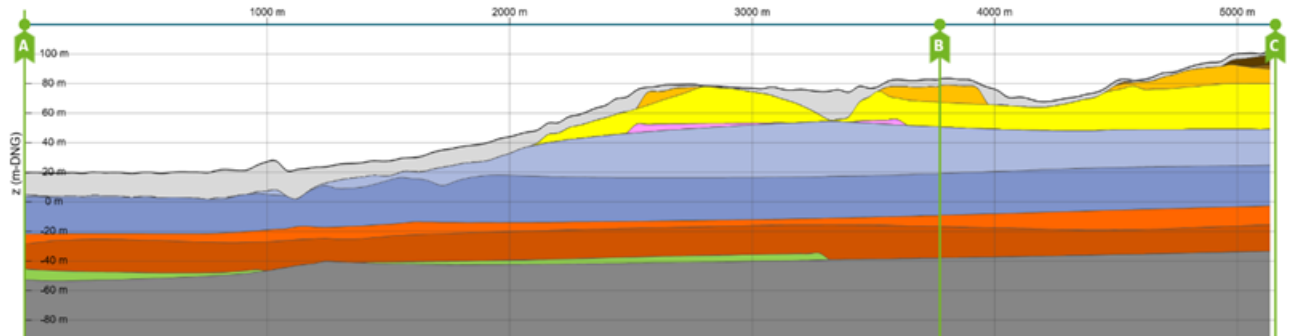


Figure 17 Cross-section of the geological condition of the upstream study area (Brugeotool)

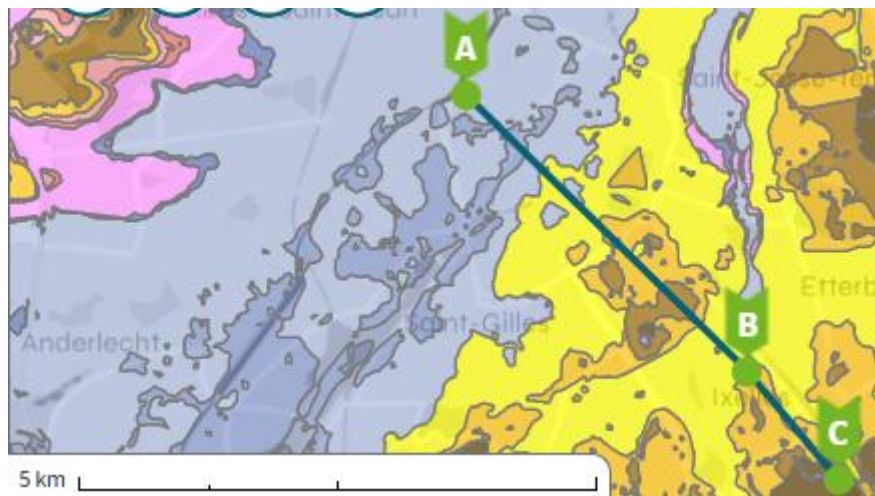


Figure 18 Cross-section of the study area seen from above (Brugeotool)

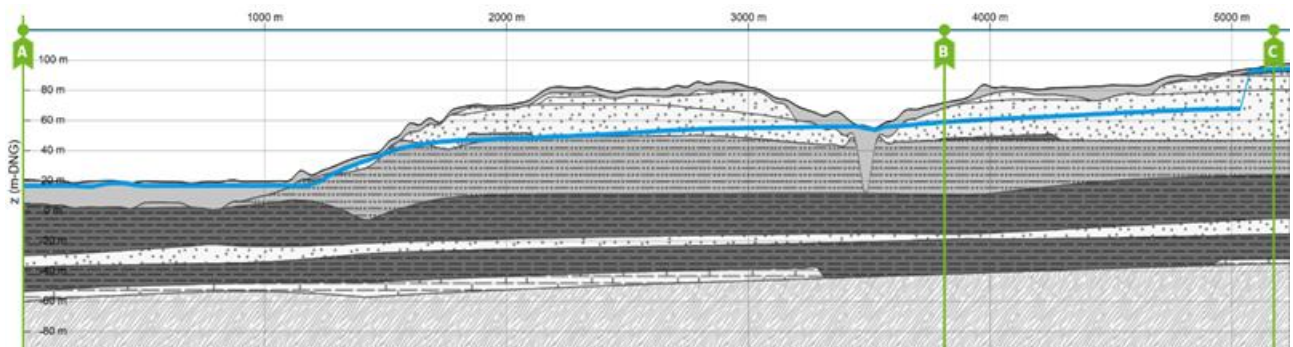


Figure 19 Cross-section of the hydrological condition of the study area (Brugeotool)

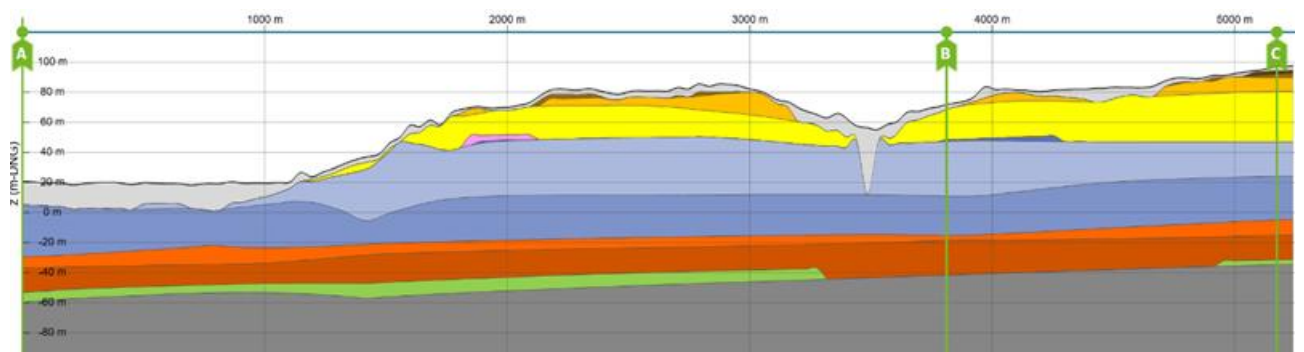


Figure 20 Cross-section of the geological condition of the study area (Brugeotool)



Figure 21 Cross-section of the downstream study area seen from above (Brugeotool)

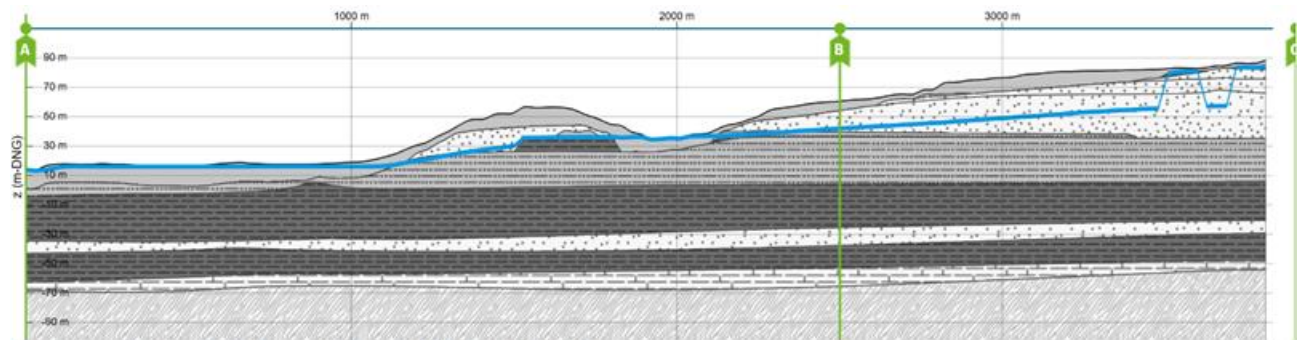


Figure 22 – Cross-section of hydrological conditions downstream study area seen from above (Brugeotool)

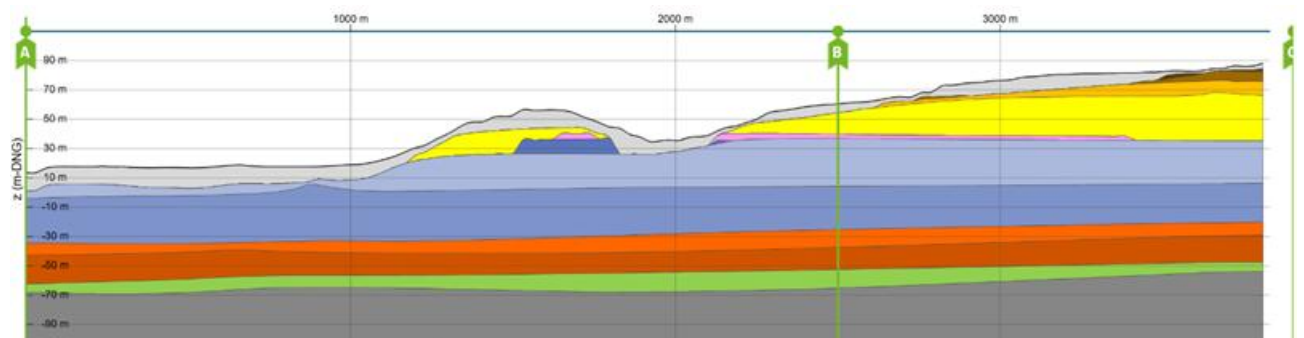


Figure 23 Cross-section of geological conditions downstream study area seen from above (Brugeotool)

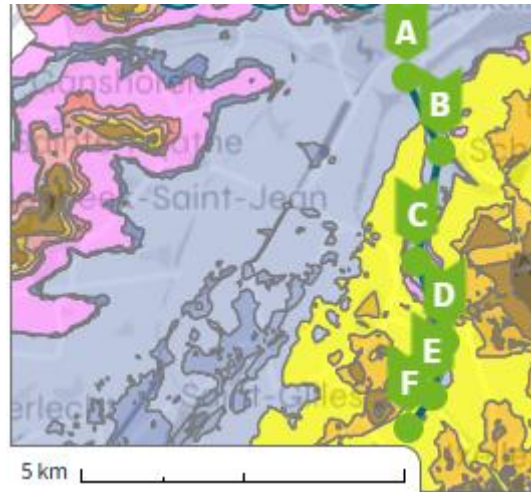


Figure 24 Cross-section along the valley seen from above (F indicates Fleegy, E indicates Rue Gray) (Brugeotool)

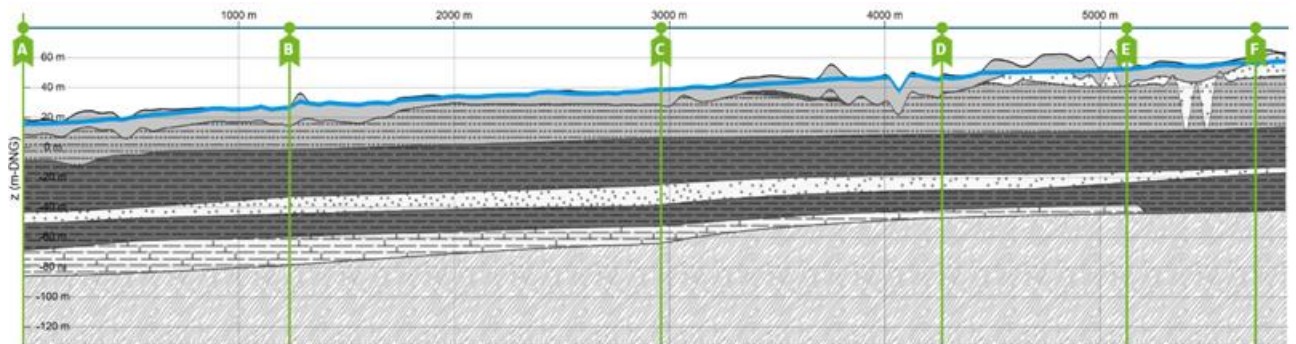


Figure 25 Cross-section of hydrological conditions along the valley profile (F indicates Fleegy, E indicates Rue Gray) (Brugeotool)

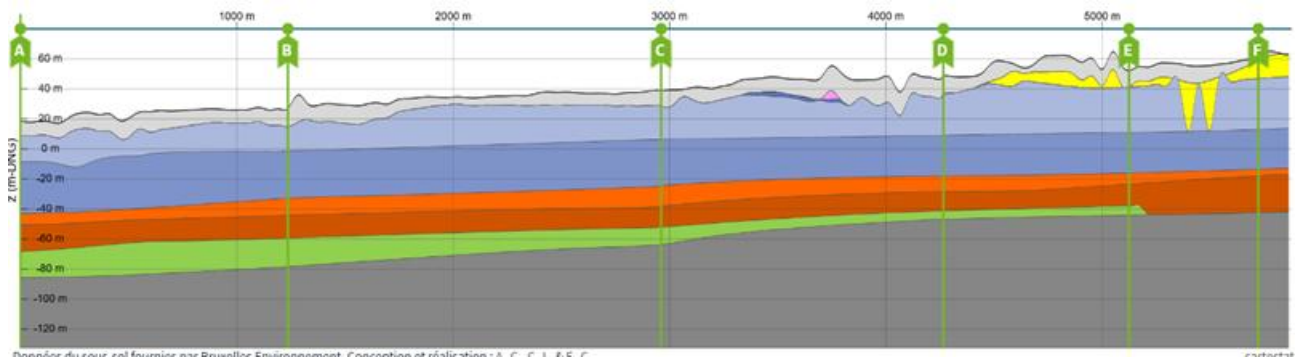


Figure 26 Cross-section of geological conditions along the valley profile (F indicates Fleegy, E indicates Rue Gray) (Brugeotool)

In order to calculate the recharge value of the rainfall event occurred in 9th July of 2024, we used the Curve Number method (CN). The values of the curve number were assigned from the HEC-HMS Technical Reference Manual tables considering the sand bank underground, according to the Cross-Sections shown before.

Table 1 Values of CN for each surface in the study area (HEC-HMS Technical Reference Manual)

Land Use and surface characteristics	Area (m ²)	CN
Agriculture	6536	81
forest	6266311	66
grass	4964275	79
buildings	14457395.7	85
paved	5356503	98
water	236390	100
roads	7971778	84

The result of the equivalent CN is 82.86, while the value of the total rainfall (42.2 mm/day) was taken from Flowbru. The calculated value of the infiltrated water, that we assume as recharge, is equal to 0.0051 m/day.

ModelMuse modelling

In this chapter, all the steps taken per run will be described. A total of five runs have been done, all changing one or multiple parameters in the model.

Scenario: Initial Steady-State Development (run 1)

1. Purpose

The objective of the initial run was to establish a baseline steady-state model for the study area based on available data for the local geological units.

2. Model parameters & Hydraulic properties

Hydraulic conductivity (K): The initial horizontal hydraulic conductivity is set at 17.28 m/d, as specified in BRUSSELS PHREATIC SYSTEM MODEL - VERSION 1.0. The vertical hydraulic conductivity is assumed to be 10% of the horizontal hydraulic conductivity.

Recharge (R): A uniform recharge rate of 0.000444 m/d (165 mm/y) across the whole model top is applied, representing the average infiltration rate for a random day in the year of 2013 as specified in BRUSSELS PHREATIC SYSTEM MODEL - VERSION 1.0.

Aquifer type: The model is configured as a confined layer for the initial calibration phase. This approach is chosen to improve the numerical stability and reduce the computational demand required by the high-resolution (10m) grid, while avoiding non-linear complexities of cell drying and re-wetting during the steady-state setup.

On top of that, due to the high urbanization of the area, it is assumed that the top of the layer can be seen as an impervious layer as well.

3. Boundary conditions & infrastructure

Senne river (RIV package): The river is implemented using the River Package to allow for seepage in both directions. The river stage is set to Model_Top-3m and the river bottom to Model_Top-4m.

Sewer system (DRN package): The urban sewer network, seen to have a groundwater baseflow, is represented using the Drain package to simulate the groundwater baseflow. The initial conductance is set to $5 \text{ m}^2/\text{d}$, to approximate the leakage in the old collector system.

External boundaries: No-flow boundaries are defined based on the topographical divide seen in the Eastern and Southern part of the research area. Boundary conditions in the North and West are defined as the Senne river boundary.

Infrastructure barriers: An impermeable wall is implemented to represent the railway, using a hydraulic conductivity of $1 * 10^{-5} \text{ m/d}$ to simulate an impermeable boundary to lateral groundwater flow. On top of that, the bridge is implemented using a hydraulic conductivity of $1 * 10^{-5} \text{ m/d}$ to simulate another impermeable boundary. Both barriers are implemented for a depth of 10 meters.

4. Observations & validation setup

Thirteen observation points are used to check the validity of the model. Eight of those points are used for calibration (points 1-8). The target head values for these points were derived from the existing Brussels steady-state model to ensure the calibration progress targets realistic groundwater elevations. Point 13 is taken from geodata.leefmilieu.brussels point "371_PZ20", whereas the other points were added manually (see Figure 27).

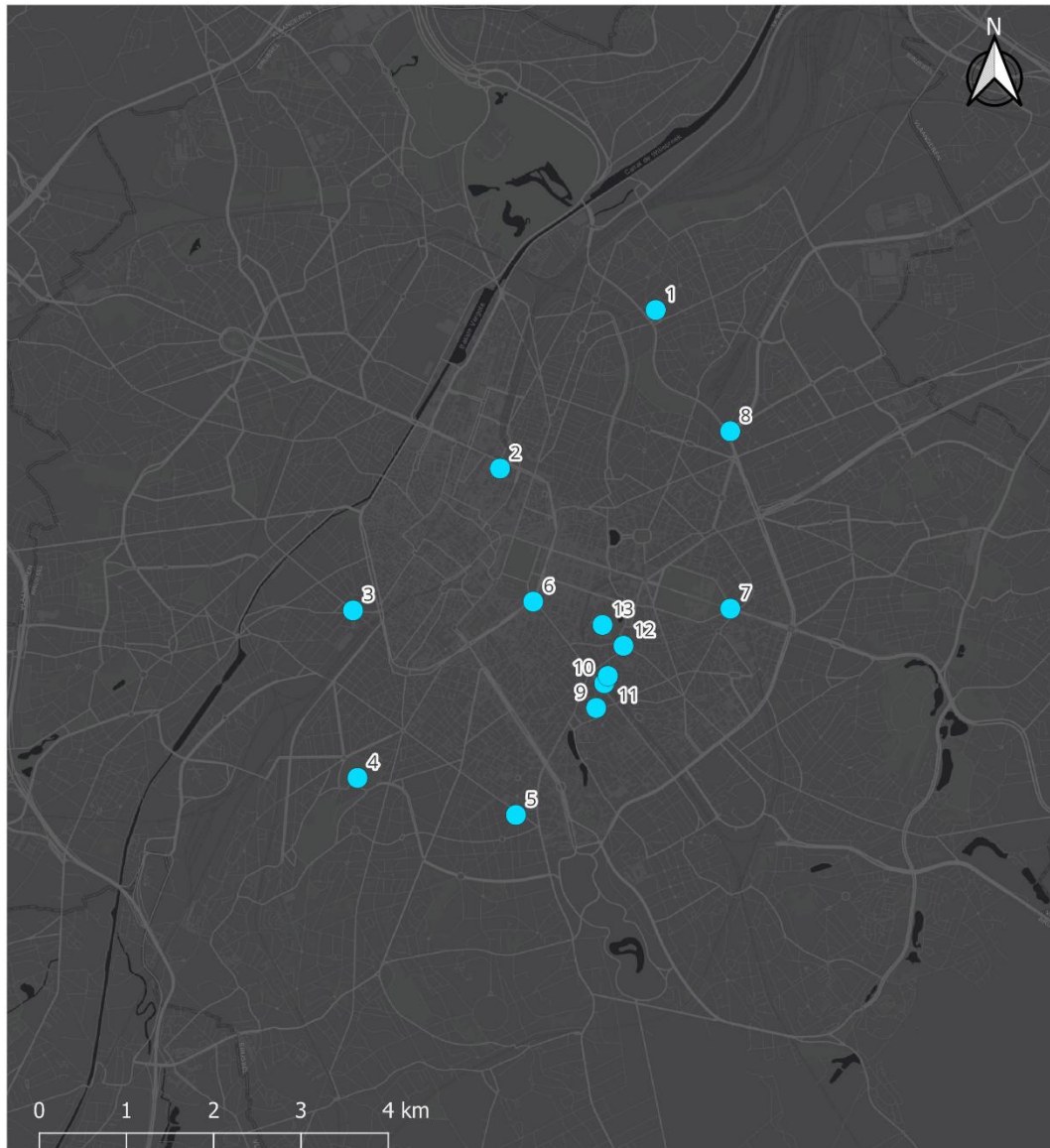


Figure 27 Observation points used in ModelMuse during all runs.

5. Results

Running the model gave the following results (see Figure 28). As can be seen, there is a very clear tendency of the groundwater to go to the Senne located on the West side of the study area. On top of that, basically no ‘forced’ groundwater flow into the Maelbeek valley is seen.

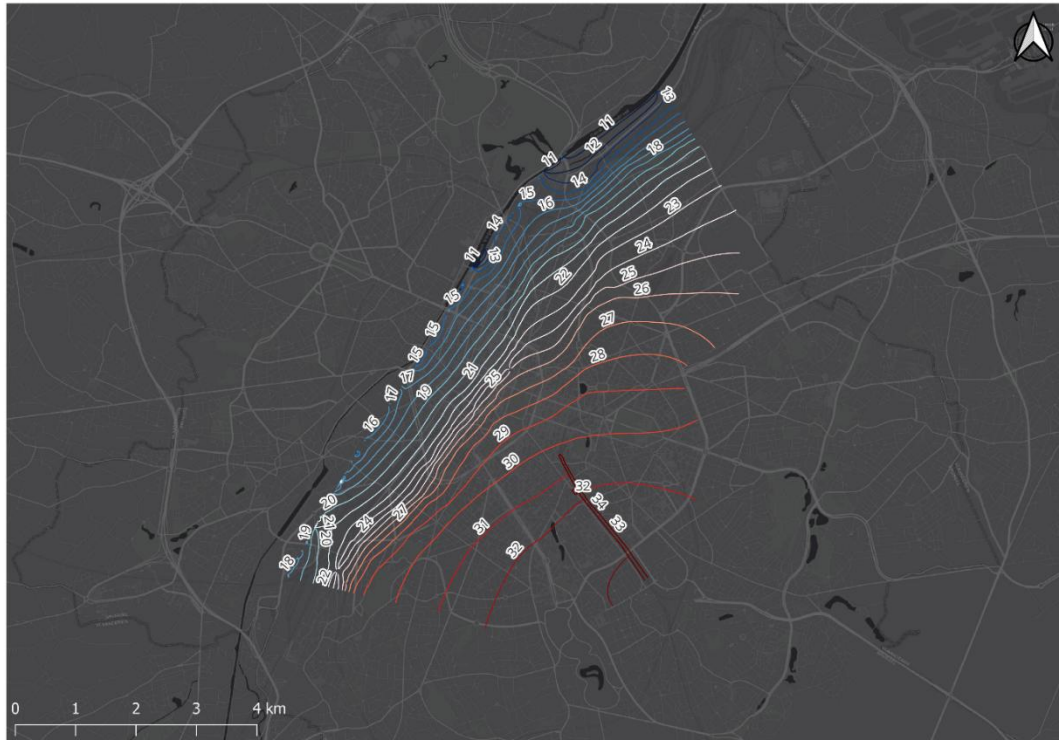


Figure 28 Contour map ModelMuse run 1.

When comparing the modelled results with the observation points, a general underestimation of the groundwater heads can be seen (see Table 2). Combined with a RMSE of 21.38 m, these results are unsatisfactory and there is a need for a calibration of the model.

Table 2 Results in observation points ModelMuse run 1.

Observation Name	Measured	Modeled	Residual
Head_obs1_1	34.46	21.730184	12.729816
Head_obs2_1	19.12	21.100532	-1.980532
Head_obs3_1	16.6	19.146749	-2.546749
Head_obs4_1	52.45	29.596552	22.853448
Head_obs5_1	63.17	32.174621	30.995379
Head_obs6_1	47.52	29.195282	18.324718
Head_obs7_1	65.71	30.13156	35.57844
Head_obs8_1	48.33	26.745503	21.584497
Head_obs9_1	53.38	31.900419	21.479581
Head_obs10_1	52.12	31.479791	20.640209
Head_obs11_1	51.86	30.87258	20.98742
Head_obs12_1	49.51	30.622764	18.887236
Head_obs13_1	49.93	26.75405	23.17595

Scenario: Calibration of initial Steady-State model (run 2)

1. Purpose

The objective of the calibration run is to develop a more realistic steady state model for the Maelbeek valley, explaining the general trend of the groundwater flow in and around the valley.

2. Model parameters & Hydraulic properties

Hydraulic conductivity (K): The initial horizontal hydraulic conductivity is set at 17.28 m/d and the initial vertical hydraulic conductivity is set as 10% of the horizontal hydraulic conductivity, as specified in BRUSSELS PHREATIC SYSTEM MODEL - VERSION 1.0. However, this value will be changed during the calibration process. For the calibration, the PEST (Parameter ESTimation) software in ModelMuse is used. This software minimizes the residuals by changing the hydraulic conductivity, ultimately resulting in a multiplication factor for K that will result in a lower RMSE.

Recharge (R): The recharge is not changed.

Aquifer type: The aquifer type is kept as confined, to avoid large computation times.

3. Boundary conditions & infrastructure

All boundary conditions and infrastructure are kept the same.

4. Observations & validation setup

Again thirteen observation points are present. However, all these points are now used for calibration purposes. The target head values for these points were derived from the existing Brussels steady-state model to ensure the calibration progress targets realistic groundwater elevations. Then, the model will be assessed based on these observation points, seeing the effect of the calibration.

5. Results

The found PEST value is “0.213942938977467”, meaning that the hydraulic conductivity should be multiplied with 0.214 to find the most accurate results. This thus results in an average horizontal K of 3.697 m/d and an average vertical K of 0.370 m/d. These new values result in the contour map as seen in Figure 29.

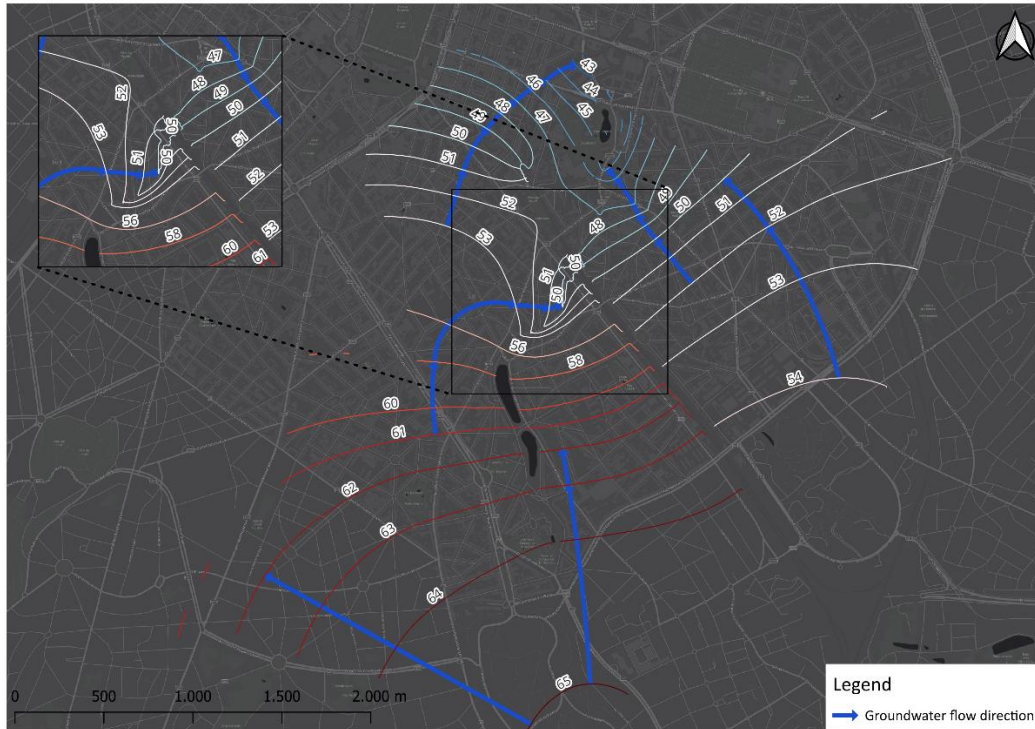


Figure 29 Contour map ModelMuse run 2. In this map, groundwater flow directions are also represented by the blue lines.

Again, when comparing the modelled results with the observation points, it can be seen that the RMSE improved drastically, going from 21.38 m in the uncalibrated model to a value of 6.25 m in the calibrated model. On top of that, when zooming in to Rue Gray (Head_obs9_1 – Head_obs11_1), the values are generally underestimated, but only by small amounts (see Table 3 and Figure 30).

Table 3 Results in observation points ModelMuse run 2.

Observation Name	Measured	Modeled	Residual
Head_obs1_1	34.46	32.024752	2.4352478
Head_obs2_1	19.12	29.468755	-10.34875
Head_obs3_1	16.6	26.009764	-9.409764
Head_obs4_1	52.45	56.733187	-4.283187
Head_obs5_1	63.17	62.981568	0.1884318
Head_obs6_1	47.52	48.756066	-1.236066
Head_obs7_1	65.71	50.523527	15.186473
Head_obs8_1	48.33	44.473281	3.8567191
Head_obs9_1	53.38	50.476877	2.9031225
Head_obs10_1	52.12	49.578999	2.5410009
Head_obs11_1	51.86	47.878028	3.9819723
Head_obs12_1	49.51	46.298875	3.2111253
Head_obs13_1	49.93	41.675856	8.2541441

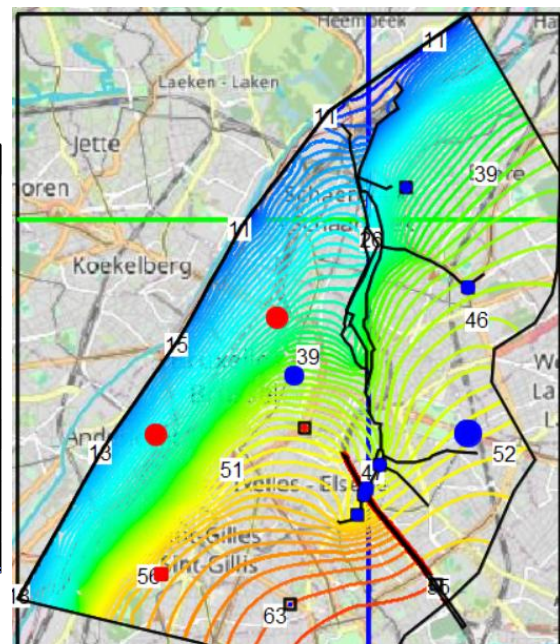


Figure 30 Output ModelMuse run 2. Blue dots represent modelling underestimation, whereas red represents modelling overestimation when compared to observed values.

Apart from the lower RMSE, the model now also shows a clear tendency of the groundwater to follow the elevation contours into Rue Gray (see Figure 29). This effect was expected as this street is the old bottom of the river Maelbeek. Generally, old river bottoms still see a high groundwater level due to the elevation contours. This model thus proves these expectations. Using the calibrated K, the model can now be tested for the event on the 9th of July 2024.

Scenario: Calibrated model on 9th of July 2024 (run 3)

1. Purpose

The objective of the third run is to see the effects on the groundwater heads during the intense rainfall event on the 9th of July 2024 and investigate whether the event caused groundwater to reach the surface in and around Rue Gray.

2. Model parameters & Hydraulic properties

Hydraulic conductivity (K): Calibrated hydraulic conductivity of scenario two is used.

Recharge (R): Recharge is calculated using the Curve Number method (CN). The values of the curve number were assigned from the HEC-HMS Technical Reference Manual tables considering the sand bank underground, according to the cross-sections extracted by Brugeotool. The result of the equivalent CN is 82.86, while the value of the total rainfall (42.2 mm/day) was taken from FlowBru, from the sensor P14 – Pluv. Flagey, located in Flagey (Ixelles). The calculated value of the infiltrated water, that we assume as recharge, is equal to 0.0051 m/day.

Aquifer type: The aquifer type is kept as confined, to avoid large computation times.

3. Boundary conditions & infrastructure

All boundary conditions and infrastructure are kept the same.

4. Observations & validation setup

Twelve observation points are used to check water levels in these specific points. However, RMSE is useless as these values are found in May 2013. The 13th observation point is compared to modelled values, as this point can be updated for the 9th of July 2024 using data from geodata.leefmilieu.brussels point “371_PZ20”.

5. Results

Running the model for the new recharge resulted in a very unrealistic contour map (see Figure 31). Water levels are found tens to hundreds of meters above the soil surface, meaning that the output of the model is not possible.

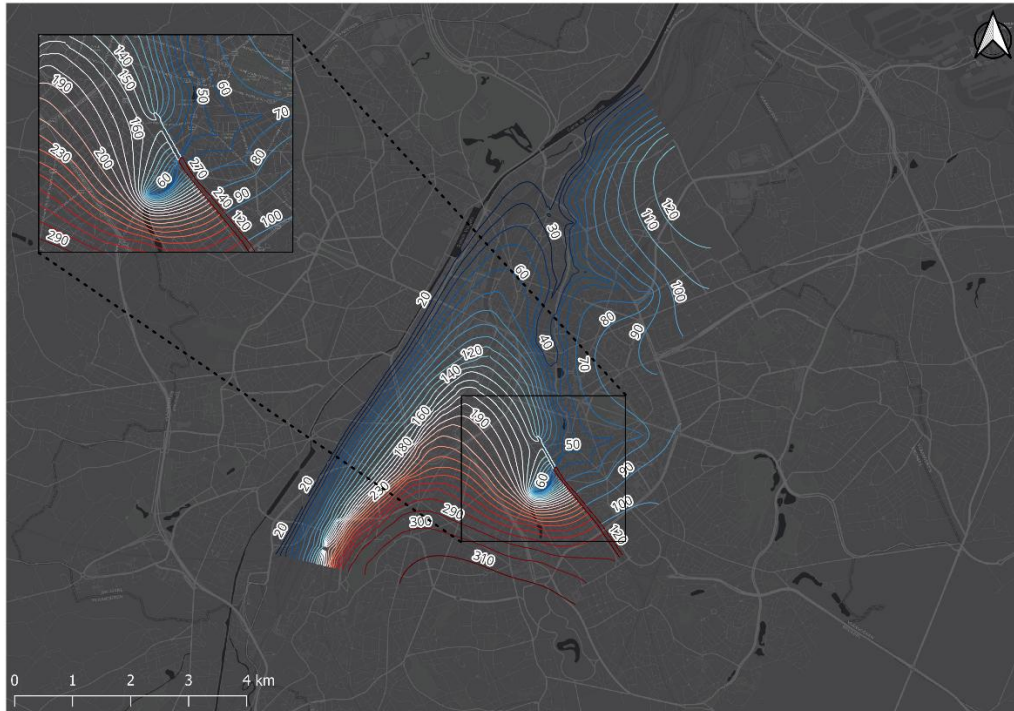


Figure 31 Contour map ModelMuse run 3.

When comparing the modelled value in point 13 with the observed value on the 9th of July 2024, it can also be seen that the value in this point is unrealistic, with a modelled value of 128.75 meters (thus 86.06 meters above the surface) while the observed value is found to be 50.03 meters in its peak. It is thus safe to say that more refinement is needed in order for the model to be used for other scenarios, especially heavy rainfall scenarios.

Scenario: Change in impermeable barriers (run 4)

1. Purpose

During the third run, it was found that the impermeable wall and bridge were blocking the flow all the way till the bottom of the model. Therefore, the conductivity of the wall will be changed and the model will be recalibrated to see the effect of the impermeable wall and bridge when they would still let some flow pass, or in other words, part of the groundwater would be able to go around the barriers.

2. Model parameters & Hydraulic properties

Hydraulic conductivity (K): Original hydraulic conductivity is used and then recalibrated using PEST software. New optimum found is 0.20161371810528, meaning that the new horizontal K is 3.484 m/d and the new vertical K is 0.348 m/d.

Recharge (R): Recharge is put back to 0.000444 m/d to allow for comparison with run 2 and run 5.

Aquifer type: The aquifer type is kept as confined, to avoid large computation times.

3. Boundary conditions & infrastructure

Infrastructure barriers: The hydraulic conductivity of both the wall and the bridge are changed to $1 * 10^{-2} m/d$ to allow some groundwater flow through the wall, allowing to model for groundwater flow around the impermeable barriers.

Other boundary conditions & infrastructure: All other boundary conditions and infrastructure are kept the same as previous runs.

4. Observations & validation setup

Thirteen observation points are used to calibrate water levels in these specific points. Then, the model will be assessed based on these observation points, seeing the effect of the calibration.

5. Results

Running the model for the lower K in the barriers resulted in higher heads downstream of the barriers when compared to run 2 (Figure 32). However, overall the results are relatively similar.

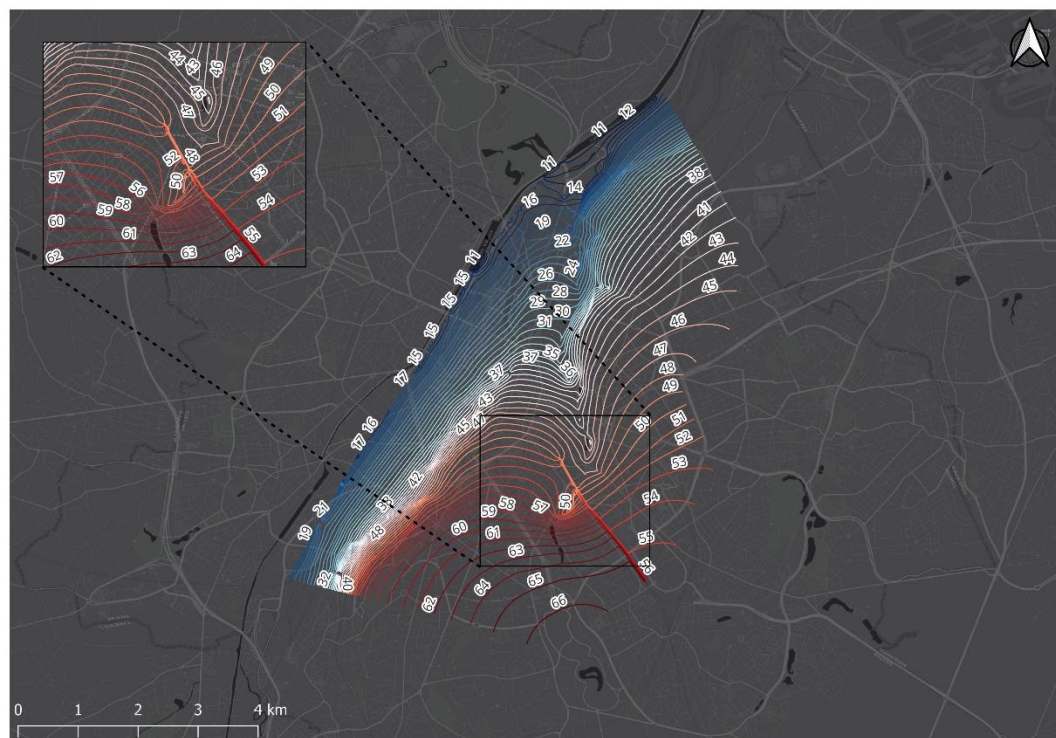


Figure 32 Contour map ModelMuse run 4.

When comparing the modelled results with the observation points, it can be seen that the RMSE improved slightly, going from 6.25 m in run 2 to a value of 6.42 m in run 4. On top of that, values are again generally underestimated when compared to observed values (Table 4 and Figure 33).

Table 4 Results in observation points ModelMuse run 4.

Observation Name	Measured	Modeled	Residual
head_obs1_1	34.46	32.39382	2.066183
head_obs2_1	19.12	29.8354	-10.7154
head_obs3_1	16.6	26.3985	-9.798498
head_obs4_1	52.45	58.16936	-5.719362
head_obs5_1	63.17	64.40977	-1.239773
head_obs6_1	47.52	49.4004	-1.880396
head_obs7_1	65.71	51.14047	14.56953
head_obs8_1	48.33	45.12397	3.206031
head_obs9_1	53.38	50.49904	2.880963
head_obs10_1	52.12	49.64512	2.474883
head_obs11_1	51.86	47.89878	3.961223
head_obs12_1	49.51	46.31461	3.195391
head_obs13_1	49.93	42.18532	7.744684

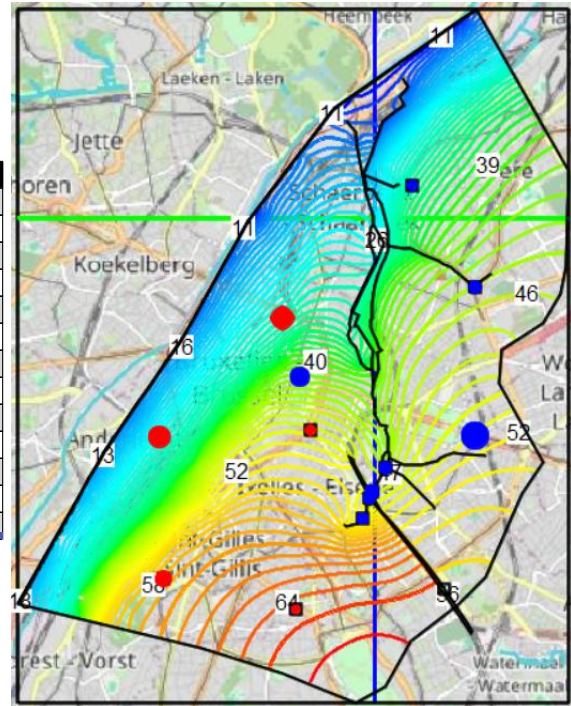


Figure 33 Output ModelMuse run 4. Blue dots represent modelling underestimation, whereas red represents modelling overestimation when compared to observed values.

Scenario: Calibration of steady state model without any barriers (run 5)

1. Purpose

To see the effect of the barriers, this run will calibrate the model for a scenario where the barriers are not present. This will show the effect of the barriers on the groundwater flow in the area.

2. Model parameters & Hydraulic properties

Hydraulic conductivity (K): Original hydraulic conductivity is used and then recalibrated using PEST software. New optimum found is 0.194236534441905, meaning that the new horizontal K is 3.356 m/d and the new vertical K is 0.336 m/d.

Recharge (R): The recharge is not changed.

Aquifer type: The aquifer type is kept as confined, to avoid large computation times.

3. Boundary conditions & infrastructure

Infrastructure barriers: Both barriers are removed. This will thus show the effects of the barriers on the groundwater flow in the area around and through Rue Gray.

Other boundary conditions & infrastructure: All other boundary conditions and infrastructure are kept the same as previous runs.

4. Observations & validation setup

Again thirteen observation points are present. All these points are used for calibration purposes. Then, the model will be assessed based on these observation points, seeing the effect of the calibration.

5. Results

Running the model after the PEST calibration gives the following contour map (see Figure 34). When comparing this run with previous runs where the barriers were present, the contour lines have a slightly different shape, but the values found for the heads are relatively similar.

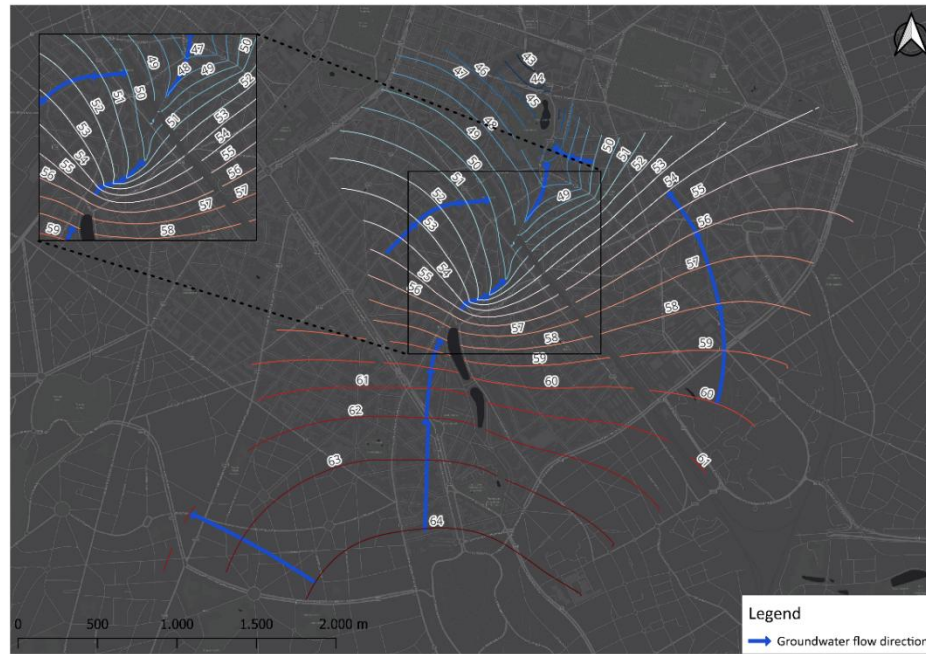


Figure 34 Contour map ModelMuse run 5.

When comparing the modelled values with the observed values (see Table 5 and Figure 35), it can also be seen that the model is relatively accurate, with a RMSE of 6.30 m.

Table 5 Results in observation points ModelMuse run 5.

Observation Name	Measured	Modeled	Residual
Head_obs1_1	34.46	32.788383	1.6716166
Head_obs2_1	19.12	29.924303	-10.8043
Head_obs3_1	16.6	26.394952	-9.794952
Head_obs4_1	52.45	57.874008	-5.424008
Head_obs5_1	63.17	63.292044	-0.122044
Head_obs6_1	47.52	49.077338	-1.557338
Head_obs7_1	65.71	52.97263	12.73737
Head_obs8_1	48.33	46.081806	2.2481943
Head_obs9_1	53.38	50.347992	3.0320078
Head_obs10_1	52.12	49.151716	2.9682845
Head_obs11_1	51.86	48.210736	3.649264
Head_obs12_1	49.51	46.363478	3.1465222
Head_obs13_1	49.93	42.192161	7.7378386

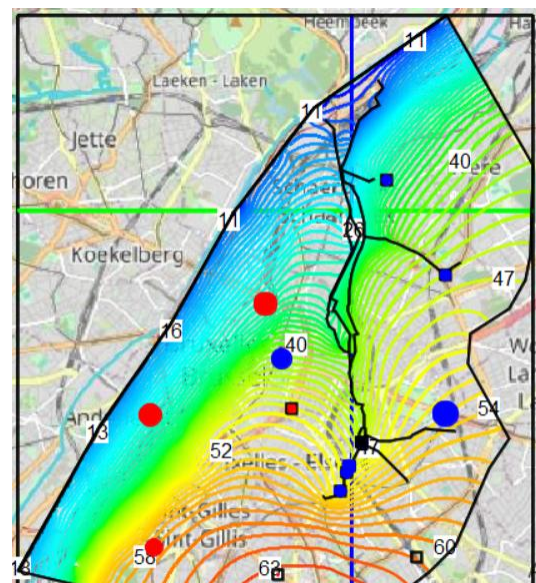


Figure 35 Output ModelMuse run 5. Blue dots represent modelling underestimation, whereas red represents modelling overestimation when compared to observed values.

Flood map

To highlight the difference between the barriers and no barriers scenarios, a flood prone area map is created. Flood-prone areas are defined as places where the groundwater is found at a maximum depth of four meters or less. Overlaying both the barriers and no barriers scenarios results in the following map (see Figure 36). As can be seen, the barriers mainly result in more flood prone areas upstream of the barriers, whereas no barriers result in higher groundwater levels downstream of the barriers.

Apart from this difference, the map also shows the confirmation of the possibility of groundwater flooding in Rue Gray. When comparing the flood-prone areas with the citizen observations during the 9th of July 2024, it can be seen that all observations have been done in the area that is highlighted as flood-prone for both the barriers and no barriers scenarios. This means that even though the barriers have an influence, this area is prone to flooding anyways, confirming the expectations of groundwater intrusion in low lying areas like basements.

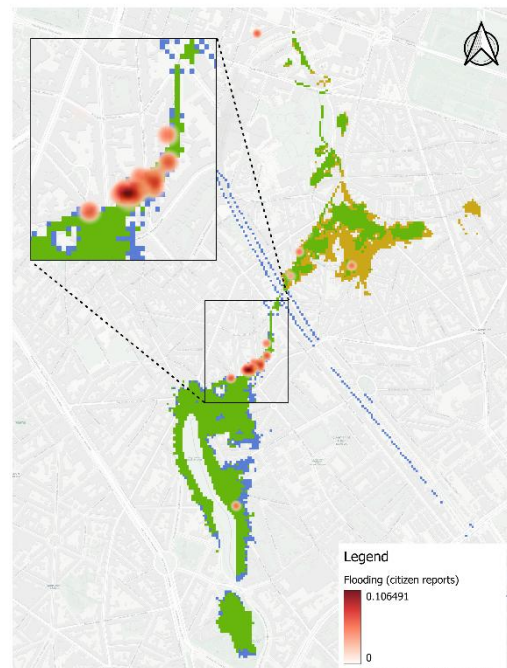


Figure 36 Map showing areas prone to flooding in the Maelbeek valley. Results based on ModelMuse outputs.

Sewer output

During the calibration phase (run 2), the idea arose to also calibrate using the sewer outflow. However, the calculated sewer drainage using FlowBru data in point “U24 – Émiss. Rive Droite (Rue du Lion)” resulted in a total outflow of 38,000 m³/d for baseflow on an average day in May 2013. When comparing this to the sewer drainage given by ModelMuse (8,299 m³/d), it was found that ModelMuse was underestimating this value. However, when checking the water balance for the model run, it was found that the total inflow is 24,963 m³, which is significantly less than the calculated base flow out of the sewer. Therefore, a calibration would not result in usable results, and this calibration step was neglected. Still, the sewer drains about 1/3 of the total inflow, while the remaining 2/3 flows to the Senne river. Therefore, it can be argued that the sewer does work as a significant drainage point, representing the conditions relatively well.

Main findings

The initial steady-state model (run 1) was insufficient for representing the study area, resulting in a RMSE of 21.38 m. By using the PEST software to adjust hydraulic conductivity, the RMSE was reduced to 6.25 m. This calibration confirmed that the initial hydraulic conductivity was too high and needed to be scaled by a factor of 0.214 to reflect realistic groundwater levels. This is most likely caused by using the assumption that the whole top layer consists out of sand, which in reality is formed out of top soil (lower K) and sand.

Even though the model was only calibrated for May 2013, the model successfully validated that groundwater flow in the region is heavily controlled by topography. The groundwater follows elevation contours, flowing toward the bottom of the Maelbeek valley. As Rue Gray is representing the old river bottom of the Maelbeek, it naturally exhibits high groundwater levels. The model confirms that groundwater tends to be shallow in this specific flood-prone section.

Then, the presence of the railway and the bridge was investigated. It was found that both the railway and the bridge act as significant hydraulic barriers, as modelling these as impermeable boundaries (run 3) clearly changed the groundwater contour shapes compared to the scenario where the barriers were removed (run 5). Besides these comparison, also a flood-prone area map was created which showed that the barriers result in a higher possibility of flooding upstream of the barriers, whereas more flooding downstream is expected in the case of no barriers. On top of that, this map proved the possibility of groundwater flooding in Rue Gray, confirming the citizen observations done during the four heavy rainfall events in 2024.

However, this model also comes with its own limitations, as the model proved not to be suited to simulate extreme weather events like the storm of July 9th 2024. This run produced groundwater levels dozens to hundreds of meters above the soil surface, which are unrealistic results. Another limitation is observation point 13. This point is using real time data to calibrate and confirm the model results. However, when writing the logbook it was found that during the importation of the point the location slightly changed, resulting in wrong values in point 13. Lastly, the model sewer output was compared to the calculated outflow and it was found that these results did not match. However, due to an insignificant amount of water coming into the model to reach these numbers, the sewer outflow was neglected.

To better understand the groundwater flow on this local scale, a multilayer model should be developed to capture the reality of the geology in more detail. On top of that, the recharge value should be checked for the initial steady state model, hopefully resulting in a match between modelled and measured sewer outflow. Finally, future models must correct the placement of observation point 13 to ensure validation data is spatially accurate.